Chinyere E. Ugwuanyi

New Brunswick, NJ | (304)-305-0259 | Email | LinkedIn | Website | GitHub

EDUCATION

Rutgers University – MBS, Analytics & Data Science | GPA: 3.89 | *May 2026*

- **Relevant Courses**: Regression, BI & Visual Analytics, Data Warehousing, Data Management, Data Structures (Java)

West Virginia State University – B.S. Chemistry | GPA: 3.84 | *April 2021*

TECHNICAL SKILLS

- **Programming & Tools:** Python (Pandas, NumPy, Scikit-learn), R, SQL/MySQL, SPSS, Qualtrics, Tableau, Excel
- **Survey & Research Operations**: Multi-campus Survey Administration, Response Rate Optimization, Reliability Analysis, Incentive Protocol Design, Knowledge Transfer Documentation
- **ML & Statistics:** Regression, Classification, Survival Analysis, Reliability Testing, Experimental Design, Longitudinal and Cohort Analysis
- **Data Engineering & Validation:** Data Cleaning, Feature Engineering, Model Validation, Quality Assurance
- **Web Technologies:** HTML/CSS

PROFESSIONAL EXPERIENCE

Graduate Research Assistant | *Rutgers - Office of Institutional Research & Decision Support (OIRDS)*
March 2025 – Present

- Led multi-campus administration of the Rutgers Undergraduate Experience Survey across NB, NWK, and CAM campuses, overseeing Qualtrics project setup, multi-wave distribution logic, response rate monitoring, and gift card lottery protocols (n = **40,000+ students**).
- Integrated and cleaned multi-source data by combining survey responses with 941 official student organization records extracted via university Engage API using R; developed text standardization, fuzzy-matching, and duplicate resolution workflows in R to ensure accurate entity linkage for reporting and incentive allocation.
- Identified and documented a Qualtrics platform anomaly affecting 950 respondents, developing corrective data workflows to maintain integrity across analysis pipelines.
- Executed large-scale roster QA and validation for seed roster files using R, Python, SPSS, and Excel to ensure accurate survey deployment across three campuses.
- Conducted **longitudinal cohort analyses** on student engagement, retention, and equity gaps to inform student success planning and strategic initiatives.
- Developed Tableau dashboards with multi-dimensional filters (college, major, degree level, campus) and executive-level reports to translate institutional data into actionable insights for administrators and academic stakeholders.
- Collaborated with academic units and student affairs departments to refine research questions, align analyses with institutional priorities, and support data-informed decision-making.
- Developed an M-index dining utilization algorithm (n = **4,800+ students**, $p < 0.001$) supporting evidence-based resource allocation; research featured in Rutgers University news publication.

Student Researcher - Computer Science Education | *Rutgers Women in Computer Science*
Spring 2024 - Fall 2024

- Designed Qualtrics surveys (n = **900+ students**) to assess Computer Science and IT&I curriculum effectiveness and student engagement.
- Built visualizations using Tableau and Plotly to support curriculum improvement recommendations.

PROJECTS

Rutgers University Student Organizations Survey Data Integration & Analysis | *Rutgers (OIRDS)*

- Integrated survey responses with official university club records by extracting institutional API data (941 organizations) and merging with respondent-provided text entries.
- Developed data-cleaning and fuzzy-matching pipelines in R to standardize club names, resolve duplicates, and link unstructured survey inputs to official organization identifiers.

- Produced a clean, analysis-ready dataset to support survey reporting, incentives distribution, and student engagement initiatives across multiple campuses.

SurveyList Webpage Redesign | *Rutgers OIRDS* — Rutgers OIRDS Survey Research Program

- Rebuilt the OIRDS Survey Research Program webpage from scratch using HTML/CSS.
- Redesigned the architecture into audience-segmented sub-pages (undergraduate, graduate, faculty, staff) with structured data tables, navigation, and accessible layout; now live at oirds.rutgers.edu.

ZIP-Code-Level COVID-19 Early Detection & Surveillance Model (NJ)

(In Progress)

- Engineer dynamic transmission indicators (velocity, acceleration, jerk, rolling variance, volatility ratios, correlation) from daily ZIP-code- and county-level COVID-19 case data using left-sided and two-sided calculus-based derivatives to capture real-time epidemic dynamics.
- Characterize epidemic wave structure by identifying inflection points and critical points in 7-day moving average curves; validate using Wilcoxon rank-sum tests comparing velocity and acceleration distributions before and after candidate dates across 21 NJ counties.
- Apply multi-method drift detection (KS test, CUSUM, Isolation Forest ensemble) to identify structural regime shifts and emerging outbreaks earlier than traditional county-level models; benchmark detection lag across waves and counties to evaluate early warning performance against established epidemiological approaches to improve targeted intervention strategies.

WiDS Datathon 2026 - Wildfire Survival Modeling Project — GitHub Repo

- Built discrete-time survival model predicting when wildfires will reach within 5km of evacuation zones; (risk horizons: 12h, 24h, 48h, 72h) to support emergency-response logistics.
- Expanded events to person-period data, engineered time encodings, trained gradient-boosted hazard model, evaluated with cross-validated Brier scores.
- Produced calibrated, time-specific risk curves to inform evacuation timing, resource staging, and operational planning under uncertainty.

Fake News Categorization – ML Pipeline (~45,000 articles) | *Accenture* — Live Demo

- Built a text classification pipeline using Python, TF-IDF, cosine similarity, and Nearest Neighbors to detect misinformation and reduce training bias and data leakage.
- Engineered 20+ linguistic features (e.g., punctuation density, sentence structure, readability scores) supporting interpretable modeling.

Airbnb NYC Price Prediction Model – Applied ML — GitHub Repo

- Built and validated regression model (40K+ listings, 5-fold CV) using Python, identifying key drivers of NYC price variability.

CERTIFICATIONS / PRESENTATIONS / ADDITIONAL INFORMATION

- **American Educational Research Association (AERA) Annual Meeting** (2026) – Co-author
  From Survey to Success: Predicting First-Year Retention Through Longitudinal Data Analysis
- **Rutgers University – Office of Institutional Research & Decision Support** - New Brunswick, NJ | *December 2025*
  Meet the Cohort of 2025: What Multi-Year New Student Survey Data Reveals About Our Newest Scarlet Knights
- **Cornell Tech | Break Through Tech AI/ML Program** - Machine Learning Foundations | *July 2025*
- **CITI Program | Social / Behavioral / Epidemiologic Research Investigators** | *June 2024*
- **Excellence in Career Readiness Award – Professionalism Category | Rutgers University |** *April 2026*